

■ 3.5.7 Integrals that *Mathematica* Can and Cannot Do

Evaluating integrals is much more difficult than evaluating derivatives. For derivatives, there is a systematic procedure involving the chain and product rules that allows one to work out any derivative. For integrals, however, there is no such systematic procedure. There are some general principles, but there are many integrals that cannot be done using these principles.

Before going any further, one must address the important question of exactly what it means to “do” an integral. As an operational matter, the important issue is usually whether one can write down a definite formula for the integral, which one can then easily manipulate or evaluate. The most useful formulae are typically ones that involve only rather simple functions, such as logarithms and exponentials.

The class of integrals that can be done in terms of “simple functions” is an important one, albeit in many respects not a particularly large one. One of the main capabilities of the built-in *Mathematica* `Integrate` function is being able to take essentially any integrand that involves only a particular set of “simple functions”, and find the integral if it can be expressed in terms of the same set of simple functions. The relevant set of “simple functions” includes rational functions, exponentials and logarithms, as well as trigonometric and inverse trigonometric functions.

Integrals of rational functions are usually quite easy to evaluate. The answers come out in terms of rational functions, together with logarithms and inverse trigonometric functions.

$$\begin{aligned} \text{In}[1] &:= \text{Integrate}[x/((x-1)(x+2)), x] \\ \text{Out}[1] &= \frac{\text{Log}[-1+x]}{3} + \frac{2 \text{Log}[2+x]}{3} \end{aligned}$$

If *Mathematica* cannot get explicit formulae for the roots of the denominator polynomial, however, it cannot give you an explicit formula for the integral.

$$\begin{aligned} \text{In}[2] &:= \text{Integrate}[1/(1+3x+x^5), x] \\ \text{Out}[2] &= \text{Integrate}\left[\frac{1}{1+3x+x^5}, x\right] \end{aligned}$$

Integrals involving logarithms and powers can be done in terms of logarithms and powers.

$$\begin{aligned} \text{In}[3] &:= \text{Integrate}[x^4 \text{Log}[x]^2, x] \\ \text{Out}[3] &= \frac{2 x^5}{125} - \frac{2 x^5 \text{Log}[x]}{25} + \frac{x^5 \text{Log}[x]^2}{5} \end{aligned}$$

Integrals of trigonometric functions usually come out in terms of other trigonometric functions.

$$\begin{aligned} \text{In}[4] &:= \text{Integrate}[\text{Sin}[x]^3 \text{Cos}[x]^2, x] \\ \text{Out}[4] &= \frac{-2 \text{Cos}[x]}{15} - \frac{\text{Cos}[x] \text{Sin}[x]^2}{15} + \frac{\text{Cos}[x] \text{Sin}[x]^4}{5} \end{aligned}$$

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When you combine “simple functions”, you sometimes get integrals that can be done.

```
In[5]:= Integrate[Sin[Log[x]], x]
Out[5]=  $\frac{-(x \cos[\log(x)])}{2} + \frac{x \sin[\log(x)]}{2}$ 
```

Often, however, you get integrals that cannot be done in terms of simple functions.

```
In[6]:= Integrate[Sin[Sin[x]], x]
Out[6]= Integrate[Sin[Sin[x]], x]
```

It is remarkable what simple integrands can lead to integrals that cannot be done, say in terms of “simple functions”. In fact, if you were randomly to combine simple functions together, most of the integrals you would get could probably not be done in terms of simple functions.

In practical usage, however, there are certain integrals that occur much more often than others. These integrals can sometimes be expressed in terms of “special functions”, which were often defined specifically as a way to represent the integrals. Most of the standard special functions are built into *Mathematica*, as discussed in Section 3.2. If you give an integral which can be done in terms of these special functions, *Mathematica* will often recognize it.

The integral $\int \frac{\log(1-x)}{x} dx$ comes out in terms of a special function known as the dilogarithm.

```
In[7]:= Integrate[Log[1-x]/x, x]
Out[7]= -PolyLog[2, x]
```

$\int \log(\log(x)) dx$ can be written in terms of the logarithmic integral function.

```
In[8]:= Integrate[Log[Log[x]], x]
Out[8]= x Log[Log[x]] - LogIntegral[x]
```

Although it may be entertaining to try and construct integrals which work out in particularly subtle ways, it is worth realizing that in practical applications, such integrals are usually quite rare. The integrals that crop up in practice can usually be divided into three classes:

- Simple integrals that can be done by simple methods.
- Integrals whose form is close to the definition of a particular special function.
- Integrals that cannot be done at all in terms of standard simple or special functions.

The rules that are built into *Mathematica* should allow you to do most of the *indefinite* integrals that are found in books of mathematical tables. Sometimes, however, you may want to add your own rules for certain kinds of integrals. Section 3.5.9 describes how to do this.

Mathematica could make the best use of rules that you add, if it could put all

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integrals into standard forms that it can recognize. The chain rule makes it possible to put all derivatives into standard forms, in terms of the *Mathematica* function **Derivative**, as discussed in Section 3.5.4. For integrals, however, there is no such systematic way to produce a standard form. Instead, *Mathematica* uses some simple transformations to try and produce easily-recognizable forms.

Mathematica makes simple transformations to try and put integrals it cannot do into standard forms.

```
In[9]:= Integrate[Sin[Sin[x]] + Cos[Cos[x]], x]
Out[9]= Integrate[Cos[Cos[x]], x] +
        Integrate[Sin[Sin[x]], x]
```

Mathematica will sometimes do part of an integral, then leave the part it cannot do in a standard form.

```
In[10]:= Integrate[(1 + Log[x]^2)/(1 - Log[x]), x]
Out[10]= Integrate[ $\frac{-2}{-1 + \text{Log}[x]}$ , x] - x Log[x]
```

There are some general transformations on integrals that *Mathematica* tries, regardless of the functions that the integrals involve.

Integrating $f'(x)$ for any f gives $f(x)$.

```
In[11]:= Integrate[f'[x], x]
Out[11]= f[x]
```

Mathematica can also deal with slightly more complicated cases.

```
In[12]:= Integrate[f'[x] f[x]^2, x]
Out[12]=  $\frac{f[x]^3}{3}$ 
```

Here is a derivative.

```
In[13]:= D[Log[Log[Log[x]]], x]
Out[13]=  $\frac{1}{x \text{Log}[x] \text{Log}[\text{Log}[x]]}$ 
```

The general transformations that *Mathematica* applies often allow it to do integrals like this.

```
In[14]:= Integrate[%, x]
Out[14]= Log[Log[Log[x]]]
```